



South Carolina

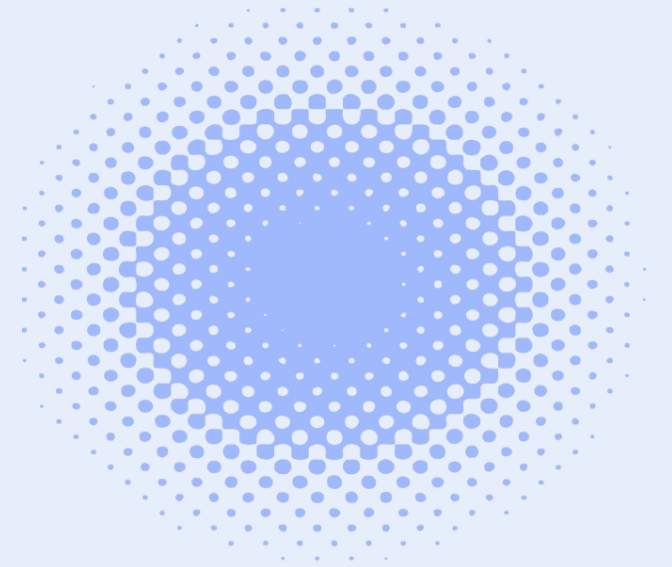
VISIT SCHEDULING • VERSION 2

Who should we offer this visit to?

A scoring rule that ranks coordinators for every open home visit

Fair rotation, family continuity, and honest workload, in one number

For lab team and PI review • August 2026



THE GOAL

Fair and efficient, at the same time

Fair

No coordinator carries a heavier share than the rest, once travel and visit length are counted honestly.

Efficient

Visits get filled quickly with someone the family already trusts, without a chain of phone calls.

FOUR THINGS THE RULE LOOKS AT



Availability

Open slot inside the date window, and no clash on the calendar.



Visit history

How many times this coordinator has already seen this family.



Workload

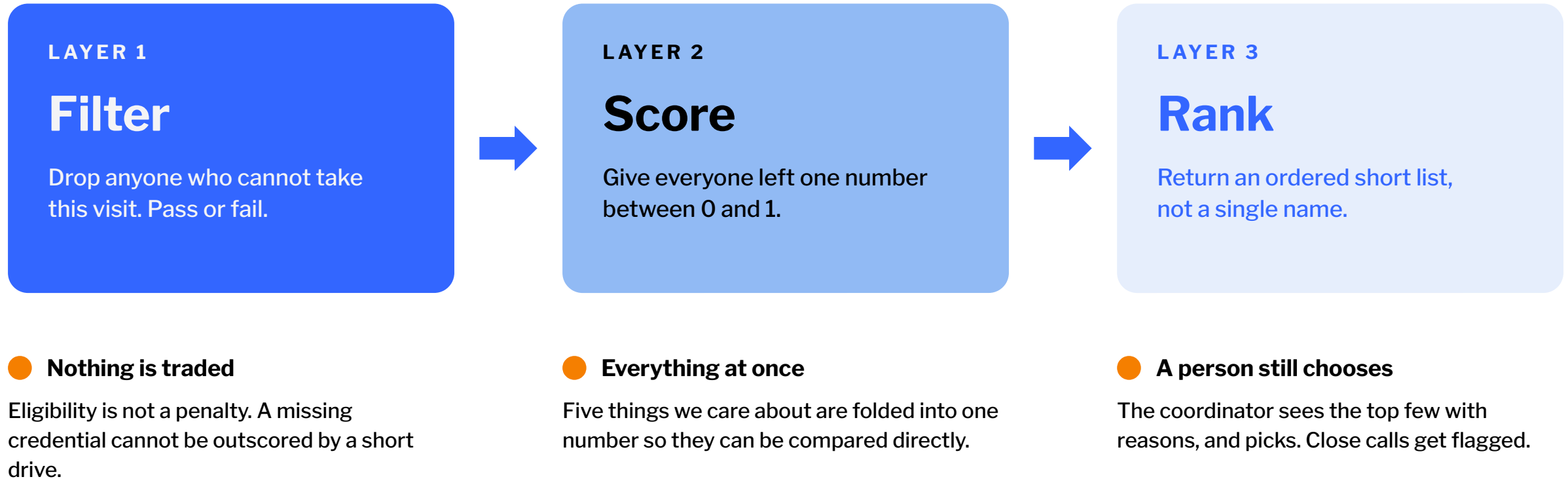
Scheduled hours this period, plus round trip travel time.



Family preference

Whether the family wants a familiar face or a fresh one.

Filter, then score, then rank



What each symbol stands for

$$i \in \mathcal{C}$$

One coordinator who could take the visit

$$v \in \mathcal{V}$$

One visit request that needs filling

$$n_{i,f}$$

Visits this coordinator has done with this family

$$k$$

How many visits it takes to feel familiar, currently 2

$$\sigma_f \in \{-1, +1\}$$

Family flag: plus one wants a familiar face, minus one wants a fresh one

$$\Delta t_i$$

Days since this coordinator's last visit of any kind

$$h_i$$

Hours already scheduled this period

$$\tau_{i,v}$$

Round trip travel time in minutes

$$\mathcal{E}_v$$

The pool of coordinators who cleared Layer 1

$$S_{i,v}$$

The final score, always between 0 and 1

$$\delta$$

Score gap below which the top two get flagged for review

$$\gamma$$

The small bonus for having done the previous checkpoint

Four eligibility checks, all must pass



Open slot

A free block inside the requested date window.



No clash

Nothing already booked over that block on their calendar.



No conflict

Not on this family's do not send list.



Credentialed

Holds every credential this study type requires.

$$\text{Eligible}(i, v) = A_{i,v} \wedge \neg X_{i,v} \wedge \neg E_{i,f} \wedge K_{i,v}$$

The wedge means and. The hooked bar means not. All four have to hold at the same time.
Availability and clash come from the calendar, so both are only as fresh as the last sync.

If nobody clears all four, the visit goes to the coordinator to reschedule. The score is never used to rescue an ineligible person.

Study type decides what a coordinator needs

$$K_{i,v} = 1 \iff \text{Req}(\text{type}(v)) \subseteq \text{Cred}(i)$$

Read it as: the visit type's required credential list has to be a subset of what this coordinator holds.

NICO

Study protocol

Required list lives in the protocol record. Nothing about it is hard coded.

NANO

Study protocol

Same lookup, different list. Adding a third study means adding a row, not editing code.

Why this matters

Nico and Nano differ on things like ADOS administration and who can take consent. Keeping that in a lookup table means the two studies never need two versions of the scoring code, and a protocol change is a data edit.

One score, five parts, new weights

$$S_{i,v} = w_1 F_{i,f} + w_2 R_{i,v} + w_3 L_i + w_4 D_{i,v} + w_5 C_{i,f}$$

$w_1 = 0.30$

Family history

How well this coordinator already knows the family

$w_2 = 0.25$

Recency

How long since their last visit of any kind

$w_3 = 0.20$

Workload

Hours already booked this period

$w_4 = 0.15$

Travel

Round trip travel time for this visit

$w_5 = 0.10$

Continuity

Bonus for having done the previous checkpoint

$$(w_1, w_2, w_3, w_4, w_5) = (0.30, 0.25, 0.20, 0.15, 0.10), \quad \sum_{k=1}^5 w_k = 1$$

$$0 \leq F, R, L, D, C \leq 1 \Rightarrow 0 \leq S_{i,v} \leq 1$$

Family history now carries the most weight, and every part is scaled to the same range, so the score itself is always between 0 and 1.

The family history term was backwards

VERSION 1

$$F_{i,f} = \sigma_f \cdot \frac{1}{1 + n_{i,f}}$$

With the flag set to plus one, meaning the family wants a familiar face, this score falls as prior visits rise. The sign and the shape disagreed, so the term did the opposite of its label.

VERSION 2

$$f_{i,f} = \frac{n_{i,f}}{n_{i,f} + k} \qquad F_{i,f} = \begin{cases} f_{i,f}, & \sigma_f = +1 \\ 1 - f_{i,f}, & \sigma_f = -1 \end{cases}$$

Familiarity climbs from 0 toward 1 as visits add up. The flag then picks the direction: keep it for a familiar face, invert it for a fresh one. No negative scores.

WHAT THE NUMBERS LOOK LIKE, WITH k = 2

Prior visits with this family	0	1	2	4	8
Familiarity	0.000	0.333	0.500	0.667	0.800
Score, family wants a familiar face	0.000	0.333	0.500	0.667	0.800
Score, family wants a fresh face	1.000	0.667	0.500	0.333	0.200

A coordinator who has never met the family now scores 0 on continuity, not 1. That was the bug worth catching before this term became the heaviest one.

Recency and workload

2

Recency

weight 0.25

$$R_{i,v} = \frac{\min(\Delta t_i, T_{\text{cap}})}{T_{\text{cap}}}$$

Days since this coordinator's last visit, capped at a ceiling such as 90 days.

Someone who worked yesterday scores near 0. Someone free for three months scores 1. This is what keeps the rotation moving.

3

Workload

weight 0.20

$$L_i = 1 - \frac{h_i}{h_{\text{max}}}$$

Hours already scheduled this period, against the busiest person on the team.

Counting hours instead of visit counts is the change here. A morning of long assessments no longer looks the same as two short check ins.

Travel and continuity

4 Travel

weight 0.15

$$D_{i,v} = 1 - \frac{\tau_{i,v}}{\tau_{\max}}$$

Round trip travel time, against the longest trip in the eligible pool.

Minutes replaced miles. Travel time and scheduled hours are now in the same unit, so the two together describe the real burden of a visit.

5 Continuity

weight 0.10

$$C_{i,f} = \begin{cases} \gamma, & \text{last checkpoint} \\ 0, & \text{otherwise} \end{cases}$$

A small fixed bonus, about 0.05, for doing the last checkpoint.

Kept deliberately small. Family history already carries continuity at weight 0.30, so this only breaks near ties in favour of the last familiar face.

Where does family preference actually live?

It is not a sixth term. It shows up in three places, each with a different amount of force.

STRONGEST

A hard override in Layer 1

A named request or an exclusion removes people from the pool outright. No score can undo it.

MIDDLE

The direction of the family history term

The flag decides whether familiarity helps or hurts, at the heaviest weight in the model.

LIGHTEST

The tie break in Layer 3

When the top two are effectively level, prior visits with the family settle it.

The reason for the split

A comfort concern is a stop sign, not a penalty. Folding it into the score would let a short drive and a light caseload quietly outvote it, which is exactly what we do not want to happen.

Three eligible coordinators, one visit

What we measure	A	B	C
Prior visits with this family	3	0	1
Familiarity, $k = 2$	0.600	0.000	0.333
Family history score	0.600	0.000	0.333
Days since last visit	5	20	40
Recency score	0.056	0.222	0.444
Hours booked this period	18	9	12
Workload score	0.000	0.500	0.333
Round trip minutes	30	75	45
Travel score	0.600	0.000	0.400
Did the last checkpoint	Yes	No	No
Continuity bonus	0.050	0.000	0.000

Illustration only. Real numbers come out of the pilot.

The setup

This family wants a familiar face, so the flag is plus one. Every score in blue is computed from the raw row directly above it.

A

The veteran

Knows the family best and lives closest, but is fully booked and worked five days ago.

B

The newcomer

Lightest caseload on the team, but has never met this family and faces the longest drive.

C

The middle

Nothing outstanding anywhere, nothing weak anywhere.

Multiply, add, then rank

COORDINATOR A

$$S_A = 0.30(0.600) + 0.25(0.056) + 0.20(0.000) + 0.15(0.600) + 0.10(0.050) = 0.289$$

RANK 2

COORDINATOR B

$$S_B = 0.30(0.000) + 0.25(0.222) + 0.20(0.500) + 0.15(0.000) + 0.10(0.000) = 0.156$$

RANK 3

COORDINATOR C

$$S_C = 0.30(0.333) + 0.25(0.444) + 0.20(0.333) + 0.15(0.400) + 0.10(0.000) = 0.338$$

RANK 1

● What this shows

C takes the top slot without leading on a single part. A has the deepest history and the shortest drive but is fully booked and worked last week, which zeroes out two terms. The gap between C and A is 0.049, inside the 0.05 review band, so the pair is flagged for a coordinator to look at rather than confirmed silently.

A ranked short list, not a single name

$$\text{Rank}_v = (i_{(1)}, i_{(2)}, \dots, i_{(m)}) \quad \text{with} \quad S_{i_{(1)}} \geq S_{i_{(2)}} \geq \dots \geq S_{i_{(m)}}$$

Offer the top K

The coordinator sees the best few with their scores and the reason each one ranked where it did. Currently K is 3.

Flag a thin margin

If the first and second are closer than the review band, the pair goes to a human before anything is confirmed.

Break a true tie

If the gap is smaller still, prior visits with the family settle it in whichever direction the family flag points.

$$S_{i_{(1)}} - S_{i_{(2)}} < \delta \implies \text{flag for coordinator review}$$

$$|S_{i,v} - S_{j,v}| < \varepsilon$$

Ranking rather than picking is the honest move. The score separates a strong candidate from a weak one reliably, but it cannot tell 0.338 apart from 0.289 in any way a person should trust blindly.

A brand new coordinator breaks three of the five parts

Part	What a new hire looks like	What the score does	Right or wrong
Family history	No prior visits anywhere	0 for a family wanting continuity	Right
Recency	No last visit on record	Undefined, or treated as a very long gap and scored 1	Wrong
Workload	Nothing booked yet	1, the maximum	Wrong
Travel	Distance is known from their address	Scores normally	Right
Continuity	Did not do the last checkpoint	0	Right

PROPOSED FIX

$$N_i < N_{\min} \implies R_i \leftarrow \text{median}_j R_j, \quad L_i \leftarrow \text{median}_j L_j$$

Until a coordinator has a handful of visits on record, borrow the team median for recency and workload instead of letting an empty record read as maximum availability. A new hire then starts in the middle of the pack and earns their own numbers, rather than being handed every visit in week one.

Filling visits one at a time leaves value on the table

Scoring one visit against one pool is greedy. Two harder questions sit behind it, and they are different problems with different tools.

ONE PERSON, ONE DAY

Which set of visits fits?

$$V(j) = \max\{ V(j-1), s_j + V(p(j)) \}$$

Pick the highest scoring set of non overlapping slots for a single coordinator. This is weighted interval scheduling, and dynamic programming does solve it: sort by end time, and for each visit take either the best answer without it or its score plus the best answer ending before it starts.

Runs in about $n \log n$ time, dominated by the sort.

WHOLE TEAM, WHOLE WEEK

Who takes which visit?

$$\max_a \sum_{v \in \mathcal{V}} S_{a(v), v} \quad \text{subject to} \quad |a^{-1}(i)| \leq c_i$$

Maximise total score across all open visits while respecting each person's capacity. This is an assignment problem, not a dynamic programming one. The standard tools are the Hungarian algorithm for one visit each, or minimum cost flow once people can take several.

I am stating a standard result here. Worth confirming against a reference before implementation.

Availability has to come from the calendar

Layer 1 cannot run on a spreadsheet of stated availability. Both checks, open slot and no clash, need live free and busy data.

Outlook, via Microsoft Graph

```
POST /users/{id}/calendar/getSchedule
```

- Least privileged permission is Calendars.Read
- Works delegated or app only with admin consent
- Up to 20 calendars per call, window under 62 days

Google Calendar

```
POST /calendar/v3/freeBusy
```

- Scope calendar.readonly is enough
- Returns busy blocks per calendar
- Needs timeMin, timeMax, and a list of calendar ids

What we get back

Busy blocks, not event details. That is all Layer 1 needs, and it keeps personal calendar content out of the system entirely.

Before anyone builds this

Endpoints and scopes were checked in August 2026 and both vendors change them. Confirm against current documentation, and settle the IRB position on reading staff calendars first.

What the pilot has to answer

1

Wire the calendar read

Get free and busy data flowing for the whole team on a read only scope. Nothing else can be tested until Layer 1 is real.

Blocked on IRB and IT

2

Replay past visits

Run the model over assignments already made and compare its ranking to what the team actually chose. Disagreements are the interesting cases.

Needs six months of history

3

Tune the weights

Adjust from the replay, not from opinion. Record the version and effective date so any past decision stays reproducible.

One afternoon with the PI

4

Debrief the surprises

Every case where the model and the team disagreed gets read out loud. Some will be model bugs, some will be undocumented rules worth writing down.

Standing item, first month

The pilot is a comparison, not a rollout. Nothing gets assigned automatically until the replay stops surprising us.

WHERE VERSION 2 LANDS

**Rank a few good options.
Let a person choose between them.**

$$\text{Offer}(v) = \begin{cases} \text{coordinator reschedules,} & \mathcal{E}_v = \emptyset \\ \text{top } K \text{ of Rank}_v, & \text{otherwise} \end{cases}$$

Rewighted

Family history now leads at 0.30, and the term behind it was rewritten so it finally matches its own label.

Honest about burden

Hours and travel minutes instead of visit counts and miles, so a long assessment day counts as one.

Still unfinished

Calendar access, new hire cold start, and joint scheduling are open. The pilot exists to close them.